Community detection in random geometric graph models: Recent advances up to 2026

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Abstract. This report surveys recent theoretical advances on community detection in random geometric graph models. These models incorporate spatial or latent geometric information and therefore generate strong local dependence, high triangle density, and transitivity. We review the main geometric community models studied in the literature, including the Geometric Block Model, the Geometric Stochastic Block Model, the Geometric Kernel Block Model, the Geometric Hidden Community Model and the Soft Geometric Block Model. We organize the literature by inference task and connectivity regime, with emphasis on weak, almost exact, and exact recovery, information-theoretic thresholds, and the algorithmic ideas used to achieve them. Particular attention is paid to the logarithmic degree regime, where recent works establish sharp exact-recovery thresholds together with efficient algorithms. We also discuss spectral approaches, extensions and open directions.

Keywords: Community detection, random geometric graphs, spatial networks, exact recovery, partial recovery, information theoretic thresholds, spectral methods

1 Introduction

Community detection studies graphs whose vertices belong to hidden groups, called communities, which influence the edge structure. Vertices in the same community tend to connect in a different way than vertices from different communities. The goal is to infer this latent community structure. This broad topic is detailed in [Fortunato, 2010]. In this report, we focus on random geometric graphs (RGG). Adding spatial information allows us to consider local information based on real or abstract proximity. In social networks, two individuals are more likely to be friends if they are geographically close, or if they share similar interests, age, etc. More concrete examples of RGG can be found in wireless networks [Gomez et al., 2015], in biology [Higham et al., 2008], and also in community detection problems [Sankararaman et al., 2020].

The main reference model for community detection is the Stochastic Block Model (SBM) [Holland et al., 1983]. Each node carries a hidden community label and edge probabilities depend on whether two nodes belong to the same community or not. This model comes from the Erdős-Rényi graph, where each node connects to others with the same probability [Erdős et al., 1960]. SBM is now well understood, see [Abbe, 2018]. Nonetheless, it has some limitations, since it is often unrealistic to assume that every pair of nodes interacts on the same probability scale. Around the same time as Erdős-Rényi, the RGG was introduced by [Gilbert, 1961]. Adding geometric information allows us to mimic some emergent behaviours of real networks, such as sparsity [Del Genio et al., 2011], or transitivity, namely the idea that a friend of a friend is more likely to be a friend.

RGG and its properties are well studied, see [Penrose, 2003] and more recently [Duchemin and De Castro, 2023]. When we add community structure, we are in a different setting. We observe the graph, and sometimes also the node locations, while the community labels remain hidden. The natural question is then whether the latent structure can be detected, and if so, how accurately the labels can be recovered. We are interested in several levels of inference. First, *distinguishability* asks whether the observed graph comes from a geometric model with communities or from a null model without community structure. This is a testing problem. It aims to distinguish the presence of a community structure, not to recover the labels. Then, if recovery is possible, one can ask how accurate it can be.

- *Weak/partial recovery*: detecting communities better than random guessing
- *Almost exact recovery*: the fraction of misclassified vertices vanishes as the number of nodes grows.
- *Exact recovery*: all community labels are correctly recovered, up to permutation.

Thus, in this report, community detection is not only understood as producing a convenient partition of the observed graph, but rather as recovering latent community labels under a probabilistic geometric model.

Previously, we emphasized properties that RGG share with real networks due to the geometric information. However, one can also obtain transitivity, and surely more, with attribute-based models [Li et al., 2018], hypergraphs [Ruggeri et al., 2023] and even geometric hypergraphs [Kovács et al., 2025]. We will not consider these models here, in order to keep the focus on the geometric aspect. A related but distinct line of work considers mixture based latent position models, such as the Gaussian Mixture Block Model [Li and Schramm, 2024], where community labels determine node positions using Gaussian distributions. See [Reynolds et al., 2009] [Rasmussen, 1999]. This differs from the geometric models considered here, where geometry directly affects edge formation rather than determining node placement. We therefore don't include these models in the core of this survey.

Moreover, among the problems mentioned above, one is *Distinguishability* which should not be confused with the *Detection Problem* that we do not study here. The latter usually asks for conditions on the parameters defining the RGG under which the model is distinguishable from an Erdős-Rényi graph. Hence, this is not directly a community question. See, [Bubeck et al., 2016] [Liu et al., 2022] [Mao et al., 2026]. Finally, at the beginning of the literature, some works were interested in recovering the latent position of nodes in the geometric space, and considered this as a community detection problem, [Valdivia and De Castro, 2019] [Eldan et al., 2022]. This is a different problem from the one we are interested in, and it's in some sense harder, because one wants to recover the position (a continuous label) of each node rather than a community (a discrete label).

2 Problem setting and inference tasks

We observe a simple undirected graph $G = (V, E)$, with no self-loops. Each vertex belongs to the ambient metric space $(\mathcal{X}, \rho)$, of dimension d, and to an unknown community. In the following, we specify each time whether this space is latent or observed. The goal is to recover this hidden label structure from the available observations, the graph and the vertex positions or solely the graph. Throughout the report, n denotes the asymptotic scaling parameter, while the total number of observed vertices is of order n. To compare models used throughout the literature and reviewed later, we introduce a unified geometric framework and remain close to the notation used in the papers. See Table 1 for an informal overview of the models and their parameters. We call it the *Unified Geometric Community Model* (UGCM) and it is defined as,

$$\mathrm{UGCM}\,(\mathcal{X}, \rho, W_n, \mu_n, Z, \pi, \mathcal{Y}, K_n)$$

where $W_n \subset \mathcal{X}$ is the window where nodes are sampled at the scale n. Then, μ_n describes how vertex locations are sampled. Usually, vertices are generated either as n i.i.d. points in W_n or as a Poisson Point Process (PPP) on W_n with the corresponding intensity measure. These two formulations lead to the same order of nodes and, in both cases, node locations are sampled independently. The Poisson formulation is often technically more convenient. We denote by Z the label set, and by $\pi = (\pi_i)_{i \in Z}$ the distribution of the labels. $\mathcal{Y}$ is the space of pairwise observation. For a standard unweighted graph one has $\mathcal{Y} = \{0, 1\}$, while for a weighted graph one typically takes $\mathcal{Y} = \mathbb{R}$. Let $\mathcal{P}(\mathcal{Y})$ denote the set of probability distributions on $\mathcal{Y}$. We consider the kernel that specifies the distribution of the pairwise observation taking account positions and labels of the two vertices, denoted by $K_n : \mathcal{X} \times \mathcal{X} \times Z \times Z \to \mathcal{P}(\mathcal{Y})$

Model	Main parameters	Spatial sampling	Latent positions?	Communities studied	Informal observation rule
GBM	$r_{\text{in}}, r_{\text{out}}, d$	n vertices i.i.d. uniform on $\mathbb{S}^{d-1}$	✓	2 balanced	Deterministic edge, hard threshold according to r_{in} and r_{out}
GSBM	$\lambda, f_{\text{in}}, f_{\text{out}}, d$	PPP on $\left[\pm\frac{n^{1/d}}{2}\right]^d$ with intensity λ	✗	2 balanced	Bernoulli edge and soft threshold according to $f_{\text{in}}(\cdot)$ and $f_{\text{out}}(\cdot)$
GKBM	λ, a, b, ϕ	PPP on $]-\frac{1}{2}, \frac{1}{2}]$ with intensity λn	✗	2 balanced	Bernoulli edge and soft threshold according to $\phi(\cdot)$, a and b
GHCM	$\lambda, \pi, d, (P_{zz'})_{z,z' \in Z}$	PPP on $\left[\pm\frac{n^{1/d}}{2}\right]^d$ with intensity λ	✗	$k \geq 2$ general	General pairwise observation law according to $P_{zz'}$
SGBM	$F_{\text{in}}, F_{\text{out}}, d$	n vertices i.i.d. uniform on $\mathbb{T}^d$	✓	$k \geq 2$ balanced	Bernoulli edge and soft threshold according to $F_{\text{in}}(\cdot)$ and $F_{\text{out}}(\cdot)$

Table 1. Overview of the geometric community models discussed in this report. For each model, the formal definition and its assumptions on the main parameters are given in the corresponding sections.

Depending on the model, the observations may consist only of the graph G or of the graph together with the node locations $X = (X_v)_{v \in V}$. Accordingly, we write the estimator as $\hat{\sigma} = \hat{\sigma}(G)$ or $\hat{\sigma} = \hat{\sigma}(G, X)$, and we view $\hat{\sigma} : V \to Z$ as an estimated labeling. We denote $\sigma^\star : V \to Z$ the ground truth community label function. Permuting the names of the communities does not change the underlying partition of the vertex set. For this reason, the quality of an estimator is measured up to permutation. Denoting by $\mathfrak{S}_Z$ the symmetric group on Z, we define

$$A_n(\hat{\sigma}, \sigma^\star) \coloneqq \frac{1}{|V|} \max_{s \in \mathfrak{S}_Z} \sum_{u \in V} \mathbf{1}\{\hat{\sigma}(u) = s(\sigma^\star(u))\}$$

Thus, $A_n(\hat{\sigma}, \sigma^\star)$ measures the fraction of correctly classified vertices after the best possible relabeling of the communities. We now give more formal definitions of the problems introduced in Section 1.

Distinguishability. Following [Sankararaman and Baccelli, 2018], distinguishability is a hypothesis testing problem. Let $(\mathbb{P}_n)_{n\geq 1}$ be the laws of a geometric random graph model with community structure and let $(\mathbb{Q}_n)_{n\geq 1}$ be the laws of a null geometric model without communities, both defined on the same observation space. Let Ω_n be the observation space. A test is a measurable function $\varphi_n : \Omega_n \to \{0,1\}$ where $\varphi_n = 1$ means that we decide in favor of the community model. We say that $(\mathbb{P}_n)$ and $(\mathbb{Q}_n)$ are distinguishable if there exist tests $(\varphi_n)_{n\geq 1}$ and a constant $\gamma > 0$ such that, under the uniform prior on $\{\mathbb{P}_n, \mathbb{Q}_n\}$,

$$\mathbb{P}\big(\varphi_n \text{ makes the correct decision}\big) = \frac{1}{2}\,\mathbb{P}_n(\varphi_n = 1)+\frac{1}{2}\,\mathbb{Q}_n(\varphi_n = 0) \geq \frac{1}{2}+\gamma-o_n(1)$$

Weak/Partial recovery. We say that weak/partial recovery is achievable if the estimator performs better than random guessing, that's to say in the symmetric case, if there exists a sequence of estimators $(\hat{\sigma}_n)_{n\geq 1}$ and a constant $\alpha > 1/k$ such that,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) \geq \alpha\big) \xrightarrow[n\to\infty]{} 1$$

Almost exact recovery. We say that almost exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n\geq 1}$ such that for every $\varepsilon > 0$,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) \geq 1-\varepsilon\big) \xrightarrow[n\to\infty]{} 1$$

Exact recovery. We say that exact recovery is achievable if there exists a sequence of estimators $(\hat{\sigma}_n)_{n\geq 1}$ such that,

$$\mathbb{P}\big(A_n(\hat{\sigma}_n, \sigma^\star) = 1\big) \xrightarrow[n\to\infty]{} 1$$

Most works in the literature focus on the case $k = 2$. Although several ideas extend beyond this setting, this is also the framework that we adopt in the following, except when specified otherwise. In that case, it is convenient to encode the labels by $\{\pm 1\} \coloneqq \{-1, +1\}$. Then "up to permutation" reduces to a global sign flip.

Let D_n denote the degree of a vertex chosen uniformly at random. We can distinguish three regimes of connectivity, depending on the growth of $\mathbb{E}(D_n)$ as $n \to \infty$. The most realistic but also the most delicate regime is the *sparse regime.* Each vertex only carries a finite amount of local information, the expected degree remains bounded,

$$\mathbb{E}(D_n) = \Theta(1)$$

In the *logarithmic regime*, the graph becomes sufficiently connected for exact recovery to become meaningful,

$$\mathbb{E}(D_n) = \Theta(\log n)$$

Finally, the *dense regime* corresponds to the easiest setting, the graph carries a lot of information,

$$\mathbb{E}(D_n) = \Theta(n)$$

3 Geometric community models and recovery thresholds

3.1 Geometric Block Model (GBM)

One of the earliest geometric models for community detection is the Geometric Block Model (GBM) introduced in [Galhotra et al., 2018]. They study the case of two balanced communities and latent positions on the unit sphere $\mathbb{S}^d$, sampled uniformly. For $1 \geq r_{\text{in}} > r_{\text{out}} \geq -1$,

$$\mathrm{GBM}(r_{\text{in}}, r_{\text{out}}, d) \coloneqq \mathrm{UGCM}(\mathbb{S}^d, \|\cdot\|_2, \mathbb{S}^d, \mathrm{Unif}(\mathbb{S}^d), \{\pm 1\}, \mathrm{Ber}(\tfrac{1}{2}), \{0,1\}, K_n)$$

with $K_n : (x, y, z, z') \mapsto \mathbf{1}\{\langle x, y\rangle \geq r_{\text{in}} \mathbf{1}_{\{z=z'\}} + r_{\text{out}} \mathbf{1}_{\{z \neq z'\}}\}$. They focus first on the one dimensional case, corresponding to $\mathbb{S}^1 \subset \mathbb{R}^2$, in the logarithmic regime,

$$r_{\text{in}} \coloneqq a \frac{\log n}{n} \qquad r_{\text{out}} \coloneqq b \frac{\log n}{n} \qquad a > b > 0$$

This regime matches the connectivity scale of one dimensional random geometric graphs. In contrast with the SBM, local motif counts are informative thanks to the transitivity of the model. Indeed, if $(u, v) \in E$ then the number of common neighbors of u and v, equivalently the number of triangles containing the edge (u, v), depends on whether u and v lie in the same community because nearby vertices tend to have overlapping neighborhoods. So triangles carry community information. In the conference version [Galhotra et al., 2018], this leads to a simple one threshold triangle counting algorithm, which already achieves exact recovery with high probability in the regime $r_{\text{in}} \geq 4 r_{\text{out}}$.

The later journal version [Galhotra et al., 2023] sharpens this analysis in two directions. First, they analyse connectivity of the random annulus graph (RAG) introduced in [Galhotra et al., 2020]. For $I_n \coloneqq \left[b \left(\frac{\log n}{n}\right)^{1/d}, a \left(\frac{\log n}{n}\right)^{1/d} \right]$ the graph $RAG_d(n, I_n)$ is obtained by connecting two latent points only when their distance belongs to I_n. In dimension 1, they prove a sharp connectivity threshold. With high probability $RAG_1(n, I_n)$ is connected if $a > 1$ and $a - b > \frac{1}{2}$ and disconnected if $a < 1$ or $a - b < \frac{1}{2}$. The condition $a > 1$ is the usual one dimensional connectivity requirement, while $a - b > 1/2$ is to get an annulus enough thick to avoid isolated vertices. For $d > 1$, [Galhotra et al., 2020] gives sufficient connectivity conditions of the same logarithmic order.

Second, [Galhotra et al., 2023] replaces the original one threshold triangle rule by a two threshold procedure. Once one leaves the regime $r_{\text{in}} \geq 4 r_{\text{out}}$, a single threshold is no longer sufficient. For inter community edges, the normalized triangle count is concentrated around $2 r_{\text{out}}$, whereas for intra community edges it depends on the latent distance and may lie either above or below that value. They therefore define two thresholds

$$E_S \coloneqq (2b + f_1) \frac{\log n}{n} \qquad E_D \coloneqq (2b - f_2) \frac{\log n}{n}$$

where $f_1, f_2 > 0$ are chosen from Chernoff type deviation bounds. An observed edge is classified as intra community if its triangle count is above E_S, as inter community if it is below E_D and is discarded when its count falls in the ambiguous interval $[E_D, E_S]$. After the filtering step, the surviving graph contains only intra community edges, but not all of them because ambiguous edges may have been deleted. Under explicit conditions on a and b, the surviving intra community graph is still connected in each community. The two connected components correspond to the two communities, this yields exact recovery.

As a consequence, [Galhotra et al., 2023] proves exact recovery in dimension 1 with explicit conditions and an impossibility result when $a < 1$ or $a - b < 1/2$. However, the exact recovery threshold for the latent one dimensional GBM is still not fully characterized, see Figure 1. A useful contrast is that this gap disappears when the latent positions are revealed. In that case, the one dimensional GBM has a sharp exact recovery threshold which coincides with the sharp connectivity threshold of the RAG. In higher dimension $d > 1$, they also prove a polynomial time exact recovery result at the correct logarithmic scale,

$$r_{\text{out}} = \Theta\left(\left(\frac{\log n}{n}\right)^{1/d}\right) \quad \text{and} \quad r_{\text{in}} - r_{\text{out}} = \Theta\left(\left(\frac{\log n}{n}\right)^{1/d}\right)$$

are sufficient, while recovery becomes impossible if one of these quantities is of smaller order [Galhotra et al., 2023].

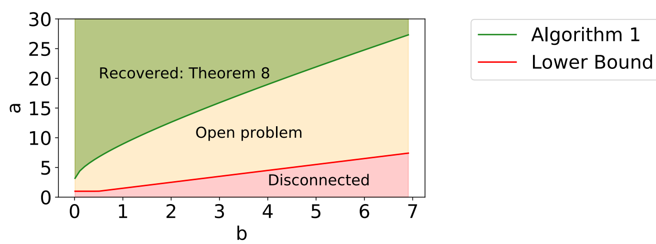

Fig. 1. Phase diagram for exact recovery in the one dimensional latent GBM in the regime $r_{\text{in}} = a \log n/n$ and $r_{\text{out}} = b \log n/n$, reproduced from Galhotra et al. [2023], Figure 5. The red region corresponds to disconnected graphs and therefore an impossibility result. The yellow region remains an open problem, and the green region corresponds to the parameter range certified by Algorithm 1 (Theorem 8).

Finally, [Chien et al., 2020] studies active learning in the latent GBM with two communities. The model is the same, but the learner is now allowed to query the labels of a small number of carefully chosen vertices to a perfect oracle. Their approach keeps the motif counting idea as a first step, in order to remove many inter community edges, and then uses label queries in a second step. They propose two algorithms. The first combines this pruning step with the graph based active learning procedure S^2 [Dasarathy et al., 2015], while the second is more aggressive and then queries one label per connected component. Their main result is that exact recovery becomes possible with high probability using only a sublinear number of queries, even in parameter regimes where the best known unsupervised guarantees fail. This gives a smooth trade off between the hardness of clustering and the query complexity. They do not prove any optimality result for this active setting. Moreover, the theoretical bounds seem somewhat pessimistic. Although the analysis favors the first algorithm, the paper notes that the second often uses fewer queries in practice, which suggests that the proof can likely be improved [Chien et al., 2020].

3.2 Geometric Stochastic Block Model (GSBM)

[Sankararaman and Baccelli, 2018] introduce the Geometric Stochastic Block Model (GSBM), which is later developed in full detail in [Abbe et al., 2021]. The model allows edge probabilities to depend jointly on distance and community labels according to $f_{\text{in}}^{(n)}(\cdot)$ and $f_{\text{out}}^{(n)}(\cdot)$. They study the case of two symmetric communities. Positions are observed and nodes belongs to $(\mathbb{R}^d, \rho_n)$ where the metric is the Euclidean distance in the sparse regime, $\rho_n := \|\cdot\|_2$, and toroidal metric in the logarithmic regime, $\rho_n := \|\cdot\|_{T_n}$. The latter is to avoid boundary effects on $W_n := \left[-\frac{n^{1/d}}{2}, \frac{n^{1/d}}{2}\right]^d$. Here, n is a scaling parameter and not the number of nodes. Points are sampled as a PPP on W_n with intensity measure $\lambda > 0$. We assume for all $r \geq 0$, $1 \geq f_{\text{in}}^{(n)}(r) \geq f_{\text{out}}^{(n)}(r) \geq 0$ and $f_{\text{in}}^{(n)} \neq f_{\text{out}}^{(n)}$ up to Lebesgue measure 0. More assumptions on these functions are given according to the regime that we are interested. The model is then defined by,

$$\text{GSBM}(f_{\text{in}}^{(n)}, f_{\text{out}}^{(n)}, \lambda, d) := \text{UGCM}(\mathbb{R}^d, \rho_n, W_n, \lambda\text{Leb}_{|W_n}, \{\pm 1\}, \text{Ber}(\tfrac{1}{2}), \{0,1\}, K_n)$$

Moreover, the kernel is defined as,

$$K_n : (x, y, z, z') \longmapsto \text{Ber}\left(f_{\text{in}}^{(n)}(\rho_n(x,y))\mathbf{1}\{z = z'\} + f_{\text{out}}^{(n)}(\rho_n(x,y))\mathbf{1}\{z \neq z'\}\right)$$

In the sparse regime, nodes have a finite expected degree. Accordingly, we assume that the connection functions do not depend on n, namely, $f_{\text{in}}^{(n)} = f_{\text{in}}$ and $f_{\text{out}}^{(n)} = f_{\text{out}}$, and that

$$\int_{\mathbb{R}^d} f_{\text{in}}(\|x\|_2) + f_{\text{out}}(\|x\|_2)\text{d}x < \infty$$

The metric is Euclidean, which allows the finite graphs to be viewed as restrictions of a single infinite random graph [Last and Penrose, 2018]. In this regime, the main question is weak recovery. By monotonicity, there exists a critical value $\lambda_c \in [0, \infty]$ separating the unsolvable and solvable phases. The first result is an impossibility criterion based on the auxiliary random connection model $H_{\lambda, g(\cdot), d}$ defined on the same PPP of intensity λ where two points at distance r are connected independently with probability $g(r)$. Denoting by $\theta(H_{\lambda, g(\cdot), d})$ the percolation probability, the probability that the connected component of a typical node is infinite. They prove that if $\theta(H_{\lambda, f_{\text{in}}-f_{\text{out}}, d}) = 0$ then weak recovery is impossible. In other words, if the informative part $f_{\text{in}} - f_{\text{out}}$ does not generate an infinite connected component, then the community information cannot propagate at long range. In particular, for $d = 1$, weak recovery is impossible for every $\lambda > 0$. To prove the lower bound, they introduce an auxiliary problem called Information Flow from Infinity and show using Palm measures, that this problem is easier than community detection. So its impossibility already implies the impossibility of weak recovery [Abbe et al., 2021]. For $d \geq 2$, this yields the explicit lower bound,

$$\lambda \leq \left(\int_{\mathbb{R}^d} \big(f_{\text{in}}(\|x\|_2) - f_{\text{out}}(\|x\|_2) \big)\, dx \right)^{-1} \quad \Longrightarrow \quad \text{weak recovery is impossible}$$

On the positive side, they introduce the Good Bad Grid (GBG) algorithm, its complexity is $O(n^2)$. Thanks to Slivnyak's theorem and the independent thinning property of the PPP, the number of common neighbors of two nearby nodes is shown to be Poisson, with two different means depending on whether the two nodes are in the same community or not. Using Chernoff bounds, this yields an exponentially small local classification error in λ. The algorithm first compares nearby pairs using such local statistics, then partitions the space into small cells and propagates consistent decisions across the grid. Its analysis relies on a coupling with a dependent site percolation process. They prove that if $d \geq 2$ then there exists $\lambda_{upper} < \infty$ such that weak recovery is solvable for all $\lambda \geq \lambda_{upper}$. Combined with the lower bound above, this implies $0 < \lambda_c < \infty$ so the sparse regime exhibits a non trivial phase transition [Abbe et al., 2021]. In some special cases, they obtain a sharp characterization of the threshold. If connection functions are hard thresholds, that is, if there exist $R_{\text{in}}, R_{\text{out}} \geq 0$ such that,

$$f_{\text{in}} : r \mapsto \mathbf{1}_{\{r \leq R_{\text{in}}\}} \qquad \text{and} \qquad f_{\text{out}} : r \mapsto \mathbf{1}_{\{r \leq R_{\text{out}}\}} \qquad \text{and} \qquad R_{\text{in}} \geq R_{\text{out}}$$

then weak recovery is solvable if and only if $\theta(H_{\lambda, f_{\text{in}}-f_{\text{out}}, d}) > 0$. So, in this hard threshold case, the phase transition is fully characterized. More generally, they also show a weak form of asymptotic optimality for GBG, when the signal is strong enough, the fraction of correctly classified vertices can be made arbitrarily close to 1. In our notation, this means that for every $\varepsilon > 0$, there exists $\lambda_0 > 0$ such that for all $\lambda \geq \lambda_0$,

$$\lim_{n \to \infty} \mathbb{P}\big(A_n(\hat{\sigma}_n^{GBG,\lambda}, \sigma^\star) \geq 1 - \varepsilon \big) = 1$$

A related question, well studied in the SBM, is whether a graph with communities can be distinguished from a null model without community structure. In [Abbe et al., 2021], the authors formulate the same testing problem for the GSBM. The null model is a random connection model without communities, with the same spatial distribution of nodes and connection function $(f_{\text{in}} + f_{\text{out}})/2$. Hence, the null and alternative models have the same average degree. They prove that, as soon as $f_{\text{in}} \neq f_{\text{out}}$ on a set of positive Lebesgue measure, the two induced probability measures are mutually singular. Therefore, the distinguishability problem is solvable with probability $1 - o_n(1)$ for every $\lambda > 0$, so it exhibits no phase transition. They construct an explicit test based on local triangle statistics. An interesting fact is that, in the sparse regime, one may detect the presence of communities while still being unable to recover a partition better than random. This is in contrast with the sparse SBM with two communities, where distinguishability and weak recovery coincide [Mossel et al., 2012].

In the logarithmic regime, the connection functions now depend on n and satisfy for $C_{\text{in}} > C_{\text{out}} \geq 0$,

$$\int_{\mathbb{R}^d} f_{\text{in}}^{(n)}(\|x\|_{T_n})\,dx = C_{\text{in}} \log n \qquad \int_{\mathbb{R}^d} f_{\text{out}}^{(n)}(\|x\|_{T_n})\,dx = C_{\text{out}} \log n$$

This is also the natural scale of connectivity for geometric graphs, which explains why exact recovery is studied in this regime. However, connectivity is not a necessary condition for exact recovery in the GSBM since the node locations are observed, isolated or weakly connected vertices may still be classified from their geometric position relative to the others. For the impossibility result, they introduce the genie estimator and the notion of a flip bad vertex, namely a vertex whose label is misclassified by the local likelihood test when all other labels are revealed. Writing N_n^{flip} for the number of such vertices, they prove that if $\mathbb{E}(N_n^{\text{flip}}) \xrightarrow[n\to\infty]{} \infty$ and if long range correlations between flip bad events are negligible, then $N_n^{\text{flip}} > 0$ with high probability. Therefore exact recovery is impossible [Abbe et al., 2021]. On the positive side, they adapt the GBG algorithm to this regime. The local pairwise tests become much more accurate, since the number of common neighbors is now of order $\log n$, and the proof no longer relies on the existence of a giant good component. Instead, they show that when the signal is strong enough, every cell is good with high probability, which yields exact recovery [Abbe et al., 2021].

They then focus on a particular logarithmic model, defined by

$$f_{\text{in}}^{(n)} : r \mapsto a\,\mathbf{1}_{\{r \leq (\log n)^{1/d}\}} \qquad f_{\text{out}}^{(n)} : r \mapsto b\,\mathbf{1}_{\{r \leq (\log n)^{1/d}\}} \qquad 0 \leq b < a \leq 1$$

This should be viewed as a non trivial instance of the logarithmic regime. Denote by ν_d the volume of the unit ball in $\mathbb{R}^d$. In this case, they obtain an explicit necessary condition for exact recovery,

$$\lambda \nu_d \Big(1 - \sqrt{ab} - \sqrt{(1-a)(1-b)}\Big) < 1 \quad \Longrightarrow \quad \text{exact recovery is impossible} \tag{1}$$

The condition < 1 means that the expected number of locally ambiguous vertices diverges. They also prove that there exists $C(a,b,d) \geq 0$ such that $\lambda\nu_d\Big(1 - \sqrt{ab} - \sqrt{(1-a)(1-b)}\Big) > C(a,b,d)$ is sufficient for exact recovery by GBG. They conjecture that the lower bound is in fact sharp. This conjecture was later confirmed by [Gaudio et al., 2024b]. The same sharp threshold also appears as a corollary of a more general model, GHCM, developed in [Gaudio et al., 2024a].

More recently, [Gaudio and Jin, 2025] extended this sharp result. They consider a model where edges are only possible if they are in a visibility radius $r(\log n)^{1/d}$ with $r > 0$, and assume mild regularity conditions on the connection functions such as support contained in $[0, r]$, a finite number of intersections between f_{in} and f_{out}, and no interval on which the two functions remain arbitrarily close. This includes in particular the previous indicator case, but also broad classes of piecewise regular or continuous kernels. Their main result is a complete characterization of the exact recovery threshold through the information quantity

$$I(f_{\text{in}}, f_{\text{out}}) := \lambda \nu_d r^d \int_0^r \left(1 - \sqrt{f_{\text{in}}(t) f_{\text{out}}(t)} - \sqrt{(1 - f_{\text{in}}(t))(1 - f_{\text{out}}(t))}\right) \frac{d\, t^{d-1}}{r^d}\, dt$$

which is a generalized Chernoff Hellinger divergence. They prove that exact recovery is achievable when $I(f_{\text{in}}, f_{\text{out}}) > 1$, while it is impossible when $I(f_{\text{in}}, f_{\text{out}}) < 1$. In dimension $d = 1$, one must additionally have $\lambda r > 1$, and exact recovery is impossible when $\lambda r < 1$. Hence, the paper shows that the exact recovery problem in the two community GSBM admits a sharp information theoretic threshold far beyond the step function case. They also provide a polynomial time algorithm achieving this threshold.

3.3 Geometric Kernel Block Model (GKBM)

[Avrachenkov et al., 2024] introduce the Geometric Kernel Block Model (GKBM) which extends the logarithmic two community geometric setting by replacing the hard threshold rule with a more general geometric kernel. The geometric positions are observed, so the space $\mathcal{X} := \left]-\frac{1}{2}, \frac{1}{2}\right]$ is not latent, and it is equipped with the toroidal metric ρ. Here again, n is a scaling parameter and not the number of nodes. Points are sampled as a PPP on $\mathcal{X}$ with intensity measure $\mu_n := \lambda n \mathrm{Leb}_{|\mathcal{X}}$ with $\lambda > 0$. They focus in the one dimensional case with two symmetric communities, although the techniques may extend to higher dimensions. The link probabilities are defined according to $\phi : \mathbb{R}^+ \to [0, 1]$ measurable with a support bounded by $r > 0$. Let $a, b \in [0, 1]$,

$$\mathrm{GKBM}(\lambda, a, b, \phi) := \mathrm{UGCM}(\mathcal{X}, \rho, \mathcal{X}, \mu_n, \{\pm 1\}, \mathrm{Ber}(1/2), \{0, 1\}, K_n)$$

Moreover they study the logarithmic regime so the kernel is set as,

$$K_n : (x, y, z, z') \longmapsto \mathrm{Ber}\left(\left(a\mathbf{1}\{z = z'\} + b\mathbf{1}\{z \neq z'\}\right)\phi\left(\frac{n}{\log n}\rho(x, y)\right)\right)$$

The hard threshold version of the logarithmic GSBM leads to the explicit bound (1). For the GKBM, the authors replace this hard threshold by a more general kernel on their restricted one dimensional model, and introduce the information quantity

$$I_\phi(a, b) := 2 \int_{\mathbb{R}_+} \left(1 - \sqrt{ab}\, \phi(x) - \sqrt{(1 - a\phi(x))(1 - b\phi(x))}\right)\, dx$$

This quantity plays the same role as in [Abbe et al., 2021]. In fact, when $\phi = \mathbf{1}_{[0,1]}$, it reduces to the one dimensional hard threshold case. Their main result shows that exact recovery is impossible if $\lambda r < 1$ or $\lambda I_\phi(a,b) < 1$. Conversely, if $\lambda r > 1$ and $\lambda I_\phi(a,b) > 1$, then exact recovery is achievable under the additional assumption that $\phi(x) > 0$ for all $x \in [0,r]$ [Avrachenkov et al., 2024]. They also propose a two phase algorithm, adapted from the recent GSBM exact recovery approach, where labels are first recovered on a small block and then propagated. The paper states that this procedure runs in time linear in the number of edges, this corresponds to a complexity of $O(n \log n)$, in the logarithmic regime [Avrachenkov et al., 2024].

3.4 Geometric Hidden Community Model (GHCM)

The most general model in this line of work is the Geometric Hidden Community Model (GHCM), as formulated in [Gaudio and Jin, 2026]. The main generalization is that pairwise observations are no longer restricted to Bernoulli edges, so we take $\mathcal{Y} := \mathbb{R}$. The geometric positions are observed, so the space $\mathcal{X} := \mathbb{R}^d$ is not latent. Let $W_n := \left[-\frac{n^{1/d}}{2}, \frac{n^{1/d}}{2}\right]^d$ equipped with the toroidal metric ρ_n. Let $k \in \mathbb{N}$, Z be a label set with $|Z| = k$ and prior $\pi := (\pi_i)_{i \in Z}$. Points are sampled as a PPP on W_n with intensity measure $\mu_n := \lambda \mathrm{Leb}_{|W_n}$ with $\lambda > 0$. The most general GHCM model, from [Gaudio and Jin, 2026], is distance dependent and is parametrized by a family of pairwise observation laws. For some $r > 0$, let

$$P = \left(P_{zz'}(t)\right)_{z,z' \in Z,\ t \in [0,r]} \in \mathcal{P}(\mathcal{Y})^{Z \times Z \times [0,r]}$$

where $P_{zz'}(t)$ is the law of the observation between two vertices of labels z and z' at distance t. Since the graph is undirected, we assume $P_{zz'}(t) = P_{z'z}(t)$. In the papers, these laws are taken either discrete or with density with respect to the Lebesgue measure on $\mathbb{R}$. We write $p_{zz'}(\cdot;t)$ for a probability mass function or a density of $P_{zz'}(t)$.

$$\mathrm{GHCM}(P, \lambda, \pi, k, d, r) := \mathrm{UGCM}(\mathcal{X}, \rho_n, W_n, \mu_n, Z, \pi, \mathcal{Y}, K_n)$$

The kernel K_n is then given by

$$K_n : (x, y, z, z') \longmapsto \begin{cases} P_{zz'}\left(\dfrac{\rho_n(x,y)}{(\log n)^{1/d}}\right) & \text{if } \rho_n(x,y) \leq r(\log n)^{1/d} \\ \delta_0 & \text{otherwise} \end{cases}$$

Thus geometry determines whether an observation is available and changes the law of this observation through the distance.

The original GHCM of [Gaudio et al., 2024a] is recovered when $P_{zz'}(t)$ does not depend on t, so geometry acts as an observation mask, it only determines whether this law is sampled or not. Hence the visibility radius is fixed to $(\log n)^{1/d}$, which corresponds to $r = 1$. This model contains the GSBM as the Bernoulli special case. The paper studies exact recovery in the logarithmic regime and identifies the information theoretic threshold,

$$\lambda \nu_d \min_{z \neq z'} D_+(\theta_z, \theta_{z'}) = 1 \qquad \text{with} \qquad \theta_z := (p_{z1}, \ldots, p_{zk})$$

with the additional one dimensional $d = 1$ obstruction, $\lambda < 1$. In the present discrete/continuous setting, D_+ is the Chernoff-Hellinger divergence given by

$$D_+(\theta_z, \theta_{z'}) = 1 - \inf_{t \in [0,1]} \sum_{r \in Z} \pi_r \int_{\mathcal{Y}} p_{zr}(y)^t p_{z'r}(y)^{1-t}\, dy$$

with the integral replaced by a sum in the discrete case. Above the threshold, exact recovery is achieved by a polynomial time algorithm in two phases under two main technical assumptions. The first phase achieves almost exact recovery using MAP estimator to initialize propagation. Then, the second phase refines the labeling and the model exhibits the same local to global amplification phenomenon as in the SBM [Abbe, 2018] Both assumptions are a bounded log-likelihood ratio, and a distinctness condition requiring that for every $z \in Z$, the laws P_{zi} and P_{zj} are different whenever $i \neq j$. The first one can however be relaxed in some special cases, in particular for Gaussian variants such as $\mathbb{Z}_2$-synchronization [Gaudio et al., 2024a]. The latter is used for the first phase during the propagation. In dimension 1, it requires the additional condition $\lambda > 1$. This extra assumption only appears when the model still has non trivial permissible relabelings, that is, when the set

$$\Omega_{\pi,P} := \{\omega : Z \to Z \text{ permutation}, \forall z, z' \in Z pi_z = \pi_{\omega(z)} \text{ and } P_{zz'} = P_{\omega(z)\omega(z')}\}$$

contains more than the identity. In the special case $|\Omega_{\pi,P}| = 1$, the paper instead uses the stronger assumption that all laws $P_{zz'}$ are pairwise distinct, and then no longer needs $\lambda > 1$.

A subsequent work removes this distinctness assumption in a relevant two community setting. In [Gaudio and Guan, 2025], they consider $k = 2$, $Z = \{1, 2\}$, $d \geq 2$, and observation laws satisfying $P_{11} \neq P_{12}$ while $P_{21} = P_{22}$. This asymmetric setting captures, for instance, geometric planted dense subgraph and geometric submatrix localization, where the goal is to identify one structured community against a homogeneous background. The information theoretic threshold is not changed. The difficulty is to build the first phase without distinctness. To resolve this, informally, the propagation of label is made by a clever and adaptive way. This yields almost exact recovery when $\pi_1 \lambda \nu_d > 1$, and then exact recovery above the usual threshold [Gaudio and Guan, 2025]. In the two community setting, the algorithm is robust to monotone semirandom perturbation, that's to say where an adversary is allowed to add intra community edges and delete inter community edges [Gaudio et al., 2024a].

The last step is done in [Gaudio and Jin, 2026], where the GHCM becomes fully distance dependent as in this definition. The information theoretic threshold then keeps the same form, but the Chernoff-Hellinger quantity must now average the information over all visible distances,

$$\lambda \nu_d r^d \min_{z \neq z'} D_+(\theta_z, \theta_{z'}; \pi, g) = 1 \qquad \text{with} \qquad g : t \mapsto \frac{d\, t^{d-1}}{r^d} \mathbf{1}_{[0,r]}(t)$$

where $\theta_z(t) := (p_{z1}(\cdot; t), \dots, p_{zk}(\cdot; t))$ and

$$D_+(\theta_z, \theta_{z'}; \pi, g) = 1 - \inf_{s \in [0,1]} \sum_{\ell \in Z} \pi_\ell \int_0^r g(t) \left(\int_{\mathcal{Y}} p_{z\ell}(y; t)^s p_{z'\ell}(y; t)^{1-s}\, dy \right) dt$$

with the inner integral over y replaced by a sum in the discrete case. The impossibility result is sharp below this threshold, with the additional one dimensional obstruction $\lambda r < 1$ when $|\Omega_{\pi,P}| \geq 2$. Above the threshold, exact recovery is again achieved by a polynomial time two phase algorithm under similar assumptions to the constant probability case.

4 Spectral methods and Soft Geometric Block Model (SGBM)

In sparse SBM, non geometric model, [Bordenave et al., 2015] showed that spectral recovery may depend on the spectral representation that is used, working with non backtracking operator rather than the common used adjacency matrix. In geometric graphs another difficulty appears, dominant eigenvectors may reflect the ambient geometry rather than the latent communities. [Méliot, 2019] analyzed the spectrum of random geometric graphs, on compact Lie groups, illustrating that geometry alone can leave a strong signature in the spectrum even in the absence of communities. A related but distinct perspective was developed by [Chen et al., 2016], who studied exact recovery from noisy pairwise measurements available only on a local observation graph, such as lines, rings, grids, or small-world networks, instead of being sampled over all node pairs. They proposed two nearly linear time algorithms, based on a spectral initialization followed by propagation and local refinement, and proved that these methods achieve the information theoretic threshold for a broad class of locality constrained graphs. Then, [Péché and Perchet, 2020] proved that spectral recovery in the SBM can remain robust under suitable random geometric perturbations. They provide the first bridge between classical spectral community detection and geometric models.

Unlike the models discussed in Section 3, [Avrachenkov et al., 2021] is included here mainly for its spectral method rather than for a new information theoretic threshold. They introduce the Soft Geometric Block Model (SGBM), where the geometric positions are latent, so the space $\mathcal{X} := \mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is not observed. The node positions are sampled i.i.d. uniformly on $\mathcal{X}$ from μ. They study two balanced communities and define

$$\mathrm{SGBM}(F_{\mathrm{in}}, F_{\mathrm{out}}, d) := \mathrm{UGCM}(\mathcal{X}, \rho, \mathcal{X}, \mathrm{Unif}(\mathcal{X}), \{\pm 1\}, \mathrm{Ber}(1/2), \{0, 1\}, K)$$

where ρ is the toroidal metric on $\mathbb{T}^d$ and

$$K : (x, y, z, z') \longmapsto \mathrm{Ber}\Big(F_{\mathrm{in}}(x-y)\mathbf{1}_{\{z=z'\}} + F_{\mathrm{out}}(x-y)\mathbf{1}_{\{z\neq z'\}}\Big)$$

Here $F_{\mathrm{in}}, F_{\mathrm{out}} : \mathbb{T}^d \to [0,1]$ are measurable connection functions. This framework contains for instance the SBM, the GBM and more.

[Avrachenkov et al., 2021] showed the Fiedler vector may correspond to a balanced geometric cut rather than to the hidden communities, so the informative eigenvector is not necessarily associated with the second eigenvalue. Rather than taking the second eigenvector, they identify the relevant eigenvalue from the Fourier coefficients of F_{in} and F_{out} and use the associated eigenvector for clustering. In the dense regime, they characterize the limiting spectrum of the normalized adjacency matrix, they prove almost exact recovery for their method and then upgrade it to exact recovery by a refinement step. Their assumptions on F_{in} and F_{out} are mild. They require a Fourier regularity condition at 0, which is satisfied for a wide class of reasonable kernels, including functions differentiable at 0 and piecewise C^1 functions continuous at 0. The main limitation is that the analysis is restricted to the dense regime, so it does not address the sparse or logarithmic settings that are central in Section 3.

[Allem et al., 2025] extend the SGBM to consider $k \geq 2$ homogeneous communities. Their goal is to identify a spectral procedure in a geometric setting with latent positions, not to find information theoretic thresholds. Writing

$$\mu_{\mathrm{in}} := \int_{\mathbb{T}^d} F_{\mathrm{in}}(x)\,dx \qquad \mu_{\mathrm{out}} := \int_{\mathbb{T}^d} F_{\mathrm{out}}(x)\,dx$$

they define the target value $\lambda^\star := n(\mu_{\mathrm{in}} - \mu_{\mathrm{out}})/k$. Their algorithm selects the $k-1$ eigenvectors of the adjacency matrix whose eigenvalues are closest to $\lambda^\star$, embeds the vertices into $\mathbb{R}^{k-1}$ using the rows of the resulting matrix, and then applies k-means. The informative object is no longer a single eigenvector but a $(k-1)$-dimensional eigenspace. In the dense regime from a Davis-Kahan perturbation argument, they prove almost exact recovery and then obtain exact recovery after a refinement step. This result is under the same assumptions as in the two-community case, namely $F_{\mathrm{in}}(0) = \widehat{F}_{\mathrm{in}}(0)$, $F_{\mathrm{out}}(0) = \widehat{F}_{\mathrm{out}}(0)$, and

$$\widehat{F}_{\mathrm{in}}(z) + (k-1)\widehat{F}_{\mathrm{out}}(z) \neq \mu_{\mathrm{in}} - \mu_{\mathrm{out}} \quad \forall z \in \mathbb{Z}^d$$
$$\widehat{F}_{\mathrm{in}}(z) - \widehat{F}_{\mathrm{out}}(z) \neq \mu_{\mathrm{in}} - \mu_{\mathrm{out}} \quad \forall z \in \mathbb{Z}^d \setminus \{0\}$$

5 Extensions and open directions

The models reviewed in this report are mostly Euclidean and are designed for inference under explicit probabilistic assumptions on labels, positions, and edge formation. A different direction comes from hyperbolic latent representations. This viewpoint is motivated by the fact that hyperbolic geometry can reproduce several structural properties frequently observed in empirical networks, such as degree heterogeneity [Barabási and Albert, 1999], strong clustering [Krioukov, 2016] and small-world [Watts and Strogatz, 1998]. In this setting, communities are not necessarily introduced as planted labels in a recovery problem, but may instead emerge from the geometric organization of the embedded graph. This idea already appears in earlier hyperbolic embedding works such as HyperMap [Papadopoulos et al., 2015], where the inferred hyperbolic coordinates reveal soft communities and can also be used for link prediction.

From this perspective, [Bruno et al., 2019] study community detection after embedding an observed graph into a latent hyperbolic space using Mercator. Communities are then extracted from the angular organization of the embedded nodes through clustering heuristics. Their goal is not to establish information theoretic thresholds or recovery guarantees, but rather to use hyperbolic geometry as an intermediate representation in which community structure may become more detectable. A limitation of this purely geometric view is that geometry alone does not necessarily explain arbitrary mesoscale mixing patterns. This point is emphasized by [Guarino et al., 2025]. They argue that classical hyperbolic models don't provide explicit control of block level interactions. To address this, they introduce a random hyperbolic block model, in which an explicit mesoscale structure is superimposed on the latent hyperbolic geometry. For the present survey, these works suggest that hyperbolic geometry is a promising representational framework, but not yet a theory of community recovery comparable to what is now available.

Returning to models reviewed above, an important open question is parameter estimation. In [Abbe et al., 2021], the authors discuss a version of the GBG procedure that does not require explicit knowledge of the connection functions. However, turning such ideas into a statistically principled estimation method remains open. In particular, estimating f_{in}, f_{out} or more generally the observation kernels, is poorly understood. Likewise, many theoretical works assume that the geometric positions are observed, whereas in practice the geometry may be latent, noisy, or only partially available.

A second direction is to weaken assumptions. For instance, in the GHCM of [Gaudio and Jin, 2026], sharp thresholds are now available in considerable generality, but the algorithmic results still rely on distinctness type assumptions ensuring that local neighborhoods contain enough information to separate the communities. Removing these assumptions, or replacing them by weaker conditions, remains an open problem.

A third direction concerns algorithms. On the one hand, higher-order spectral clustering is effective in the dense regime [Avrachenkov et al., 2021, Allem et al., 2025]. On the other hand, the sharp exact recovery results in the logarithmic regime rely on local propagation followed by refinement [Galhotra et al., 2023, Abbe et al., 2021, Avrachenkov et al., 2024, Gaudio and Jin, 2026]. A natural question is therefore whether one can design spectral or hybrid methods that remain effective in sparser geometric graphs.

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